

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028

GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, surfacing **86,387** duplicate records and a **47%** timestamp null rate to establish the correct data source for downstream ML monitoring.
- Engineered a time-series ML anomaly detector modeling seasonality across **30+** countries.
- Built SQL and MLOps pipelines automating weekly retraining and hourly ML inference.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Implemented asynchronous batch processing and storage workflows in GCP, yielding a **35%** increase in execution speed and a **60%** reduction in redundant database write operations.
- Engineered a Flask backend unifying **100k+** datapoints from **20+** heterogeneous device sources.
- Automated environment setup and data seeding via Bash on Linux, cutting testing time by **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy data pipelines for a **15%** capacity boost across **1M+** weekly events.
- Built advanced data structures for analytics modules, boosting reporting speed by **20%**.
- Refined Panels-based UI through iterative user testing, lifting workflow efficiency by **15%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), building weight-matrix extraction utilities that reduced **500GB** checkpoints to **13GB** manifolds.
- Optimized architectural placement scripts and preprocessing for a **30x** runtime reduction (30 min to <1 min), enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.
- Applied **Rent's Rule** and Louvain clustering to raise ICN density to **13.25%** in Llama-2-70B.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Gemini API, MCP, Django, React, SQLite*

- Architected multi-agent workflows cutting hallucinations by **80%** with an **83% faster** runtime.
- Engineered custom **MCP** servers to standardize tool access across the agent framework.
- Developed **7 REST API endpoints** and prototypes validated by **15+** chess players and coaches.

PROJECTS

NanoGPT | *Python, PyTorch, Transformers, Neural Networks*

Mar. 2026

- Built a transformer from scratch in **PyTorch** with multi-headed attention and backprop.
- Engineered dataloaders streaming tokenized datasets at **100%** GPU utilization while training.
- Trained a character-level chatbot with **100k+** parameters end-to-end in **Python**.

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite*

April 2026

- Developed a concurrent CLI with **Asyncio** for real-time URL blacklisting and Notion API sync.
- Optimized state persistence via **aiosqlite** for non-blocking, concurrent database operations.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, TypeScript, SQL (Postgres/SQLite), JavaScript, Bash

Development: React, Django, Node.js, FastAPI, Flask, Asyncio, WebSockets, JUnit, PyTest

AI & Systems: PyTorch, TensorFlow, LLMs, MCP, AWS, GCP, Docker, GitHub Actions, Linux/Unix